

ANDRES BODINGTON

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HIGHLIGHTS OF QUALIFICATIONS

- High-Achieving Computer Science student (GPA 4.0/4.0) with a strong foundation in Backend Development, Object-Oriented Programming, and Database Design
- Software Engineering Intern at Andrómeda Ventures, cutting LLM API costs on production chatbots via real-time rate limiting, WebSocket cost tracking, and prompt optimization
- Proficient in Java, C++, Python, and TypeScript, with proven experience designing a multi-tenant REST API (AquaAnalytics) that automated data ingestion and managed 10,000+ race results
- Demonstrated discipline, leadership, and teamwork as a Second Team Scholar All-America swimmer and active member of the AI/Data Science and Cybersecurity clubs
- Bilingual in Spanish and English; semi-fluent in Portuguese, with strong cross-cultural communication skills

SUMMARY OF TECHNICAL QUALIFICATIONS

Operating Systems: Windows, Linux, macOS, VMWare

Networking: LAN, WAN, Cloud, and Wireless system design, configuration, and troubleshooting

Programming Languages: C++, Java, Python, TypeScript, SQL, JavaScript

Frameworks & Web Technologies: NestJS, Spring Boot, LangChain, REST APIs, WebSockets, HTML5, CSS3, JSON

Databases: PostgreSQL, SQL

Development Management Tools: Git, GitHub, unit testing

Content Creation Tools: Adobe Photoshop, Adobe Illustrator, Microsoft Office

Skills: Backend Development, Object-Oriented Programming (OOP), Data Structures and Algorithms, Database Design, Software Engineering, AI and Machine Learning, AI Agent Orchestration, LLM Cost Optimization

EXPERIENCE

Andrómeda Ventures — Software Engineering Intern

June 2026 – August 2026

- Optimized the AI backend powering the Mandarin Tec and Tribu Deportiva production chatbots, cutting LLM API spend with no loss in response quality or added latency
- Built a real-time rate limiter enforcing a \$0.10 hard cap per user, with automated alerts at 80% of budget
- Migrated the OpenRouter integration to WebSockets, enabling zero-latency, real-time spend tracking per request
- Researched prompt structure and API configuration to lower token usage per conversation while preserving answer accuracy
- Developed custom LLM tools, including a soccer player comparison tool for Tribu Deportiva, delivering structured answers beyond generic chatbot replies
- Evaluated AI agent frameworks and implemented a LangChain-based architecture that reduced hallucinations by over 70%
- Refactored nested logic into the Repository Pattern with custom NestJS wrappers, centralizing error handling with full unit-test coverage

PROJECT

AquaAnalytics – Swim Performance Analytics REST API (Java, Spring Boot, PostgreSQL)

- Built a multi-tenant REST API enabling swim coaches to track athlete performance, analyze race data, and calculate improvement metrics across 50+ swimmers and 200+ events
- Automated meet results ingestion by parsing PDFs with Apache PDFBox, reducing manual data entry time by 90% while maintaining accuracy through name matching
- Implemented advanced analytics endpoints calculating improvement rates, fatigue indicators, and split consistency scores using statistical algorithms on 10,000+ race results

EDUCATION

Lewis University, Romeoville, IL

August 2025 – Present

Bachelor of Science in Computer Science, GPA: 4.0/4.0

Mars Hill University, Mars Hill, NC

August 2023 – May 2025

Bachelor of Science in Computer Science, GPA: 3.96/4.0

Dean's List (Fall 2023, Spring 2024, Fall 2024, Spring 2025)

HONORS AND AWARDS

- Recipient, Merit Transfer Scholarship for academic achievement, 2025–Present; Scholastic All-America 2024
- Second Team Scholar All-America 2025
- Eight-time conference finalist and Mars Hill University school record holder (500, 1000, 1650 Freestyle)

LEADERSHIP, ACTIVITIES, AND VOLUNTEER EXPERIENCE

- Member, Lewis University AI and Data Science Club and Cybersecurity Club (August 2025 – Present)
- Member, Mars Hill University Men's Swim Team (2023–2025); Lewis University Men's Swim Team (2025–Present)
- Red Cross Volunteer, Marshall, NC (September–October 2024): Assisted in Hurricane Helene's recovery efforts